I.C.S.E

GRADE-10

Computer applications project

Topic :Currency converter

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Acknowledgement

In the accomplishment of this project successfully, many people have bestowed upon us their blessings and heart pledged support, this time we are utilizing to thank all the people who have been concerned with this project.

Primarily we would like to express our special thanks of gratitude to our computer teacher, Mr. Prakash, who gave us a golden opportunity to do this project and also provided support in completing the project. We would also like to thank our parents for supporting us and guiding us.

Last but not the least we would like to thank all our teammates for co-operating to finish this project.

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SYNOPSIS

In this currency converter program users are provided with an option to select the type of conversion, i.e. from “this” currency to “that” currency. This simple program allows users to enter the amount to be converted (say currency in Dollars), and display the converted amount (say currency in Euro).

This program is used to convert one country’s currency to another. It has a total of 51 countries .There are a total of 2061 combinations available for currency conversion.
The program first accepts the name of the country whose currency has to be converted,and then the country to which currency has to be converted to (destination country). The amount is then accepted which is first converted to Indian currency that is Indian rupee. Indian rupee is converted to the destination currency. In this way the final value is displayed. The list of countries is stored in an array and in the same order the value of currency of those countries in Indian rupees is stored in another array named “currencytoIndian”. Another array contains the inverse conversion of a currency in Indian rupee in the same position as countries.

This program can be used by any user, but it is mainly useful for business, shares, and finance related areas where money transfer and currency exchange takes place on a daily basis.

SOURCE CODE

```
import java.util.Scanner;//java package
class CurrencyConversion
{
    public static void main(String args[])//void main
    {

        String countryname[]= new String[51];//declaring and initializing array
        //Initialzing array elements
        countryname[0]="Rwanda";
        countryname[1]="Somalia";
        countryname[2]="Sri Lanka";
        countryname[3]="Sudan";
        countryname[4]="Thailand";
        countryname[5]="Tonga";
        countryname[6]="Pakistan";
        countryname[7]="Uganda";
        countryname[8]="Uzbekistan";
        countryname[9]="Uruguay";
        countryname[10]="India";
        countryname[11]="Singapore";
        countryname[12]="Usa";
        countryname[13]="Canada";
```

```
countryname[14]="Denmark";
countryname[15]="UK";
countryname[16]="Japan";
countryname[17]="China";
countryname[18]="Holland";
countryname[19]="Italy";
countryname[20]="Sweden";
countryname[21]="Malaysia";
countryname[22]="Mexico";
countryname[23]="Thailand";
countryname[24]="Germany";
countryname[25]="Turkey";
countryname[26]="Spain";
countryname[27]="Portugal";
countryname[28]="Indonesia";
countryname[29]="Venezuela";
countryname[30]="Russia";
countryname[31]="Afganistan";
countryname[32]="Nepal";
countryname[33]="Australia";
countryname[34]="New Zeland";
countryname[35]="South Korea";
countryname[36]="Ukraine";
countryname[37]="Bangladesh";
countryname[38]="Brazil";
countryname[39]="Taiwan";
countryname[40]="UAE";
countryname[41]="Saudi Arabia";
countryname[42]="Turkey";
countryname[43]="Israel";
```

```
countryname[44]="Sudan";
countryname[45]="France";
countryname[46]="Poland";
countryname[47]="Phillipines";
countryname[48]="European Union";
countryname[49]="Fiji";
countryname[50]="Georgia";

String currencyname[]= new String[51];//declaring and initializing array
//Initializing array elements
currencyname[0]="Rwandan franc";
currencyname[1]="Somali shilling";
currencyname[2]="Sri Lankan rupee";
currencyname[3]="Sudanese pound";
currencyname[4]="Thai baht";
currencyname[5]="Tongan paanga";
currencyname[6]="Pakistani rupee";
currencyname[7]="Ugandan Shilling";
currencyname[8]="Uzbekistani som";
currencyname[9]="Uruguayan peso";
currencyname[10]="Rupees";
currencyname[11]="Singapore dollar";
currencyname[12]="American dollar";
currencyname[13]="Canadian Dollar";
currencyname[14]="Danish Krone";
currencyname[15]="Pound Sterling";
currencyname[16]="Japanese Yen";
currencyname[17]="Chinese Yuan";
currencyname[18]="Euro";
currencyname[19]="Euro";
```

```
currencyname[20]="Malaysian Ringgit";
currencyname[21]="Mexican Peso";
currencyname[22]="Thai Baht";
currencyname[23]="Euro";
currencyname[24]="Euro";
currencyname[25]="Turkish lira";
currencyname[26]="Euro";
currencyname[27]="Indonesian Rupiah";
currencyname[28]="Venezuelan Boliviar";
currencyname[29]="Russian Ruble";
currencyname[30]="Swedish Krona";
currencyname[31]="Afghan Afghani";
currencyname[32]="Nepalese Rupee";
currencyname[33]="Australian Dollar";
currencyname[34]="New Zealand Dollar";
currencyname[35]="South Korean Won";
currencyname[36]="Ukrainian Hryvnia";
currencyname[37]="Bangladeshi Taka";
currencyname[38]="Brazilian Real";
currencyname[39]="New Taiwan Dollar";
currencyname[40]="United Arab Emirates Dirham";
currencyname[41]="Saudi Riyal";
currencyname[42]="Turkish Lira";
currencyname[43]="Israeli New Sheqel";
currencyname[44]="South Sudanese Pound";
currencyname[45]="French Franc";
currencyname[46]="Polish Zloty";
currencyname[47]="Philippine Peso";
currencyname[48]="Euro";
currencyname[49]="Fijian Dollar";
```

```
currencyname[50]="Georgian Lari";

double currencytoIndian[]=new double[51];//declaring and initializing array
 //Initialzing array elements
currencytoIndian[0]=0.0755233;
currencytoIndian[1]=0.1435503;
currencytoIndian[2]=0.2220;
currencytoIndian[3]=0.1433;
currencytoIndian[4]=2.25940;
currencytoIndian[5]=34.312;
currencytoIndian[6]=0.363741;
currencytoIndian[7]=0.0217;
currencytoIndian[8]=0.0072811;
currencytoIndian[9]=2.0499122;
currencytoIndian[10]=1.0;
currencytoIndian[11]=59.19132;
currencytoIndian[12]=81.65501;
currencytoIndian[13]=60.89386;
currencytoIndian[14]=11.269866;
currencytoIndian[15]=96.91598;
currencytoIndian[16]=0.577516;
currencytoIndian[17]=11.432435;
currencytoIndian[18]=83.81707;
currencytoIndian[19]=83.81707;
currencytoIndian[20]=17.828605;
currencytoIndian[21]=4.175964;
currencytoIndian[22]=2.2594082;
currencytoIndian[23]=83.81707;
currencytoIndian[24]=4.3872;
currencytoIndian[25]=78.85;
```

```
currencytoIndian[26]=83.81707;
currencytoIndian[27]=0.005208;
currencytoIndian[28]=0.0032854;
currencytoIndian[29]=1.3459987;
currencytoIndian[30]=7.629566;
currencytoIndian[31]=0.922406;
currencytoIndian[32]=0.624926;
currencytoIndian[33]=54.09264;
currencytoIndian[34]=50.15578;
currencytoIndian[35]=0.060247;
currencytoIndian[36]=2.2105764;
currencytoIndian[37]=0.79686;
currencytoIndian[38]=15.280613;
currencytoIndian[39]=2.6213547;
currencytoIndian[40]=22.231149;
currencytoIndian[41]=21.734117;
currencytoIndian[42]=4.386892;
currencytoIndian[43]=23.864771;
currencytoIndian[44]=0.626976;
currencytoIndian[45]=12.9647;
currencytoIndian[46]=18.102612;
currencytoIndian[47]= 1.440386;
currencytoIndian[48]=85.05509;
currencytoIndian[49]=36.88877;
currencytoIndian[50]=29.970606;

double inverseconversion[]=new double[51];//declaring and initializing array
//Initialzing array elements
inverseconversion[0]=13.240;
```

```
inverseconversion[1]=6.9662;
inverseconversion[2]=4.502;
inverseconversion[3]=6.9744;
inverseconversion[4]=0.4425;
inverseconversion[5]=0.0291;
inverseconversion[6]=2.7492;
inverseconversion[7]=45.888;
inverseconversion[8]=137.34137;
inverseconversion[9]=0.487826;
inverseconversion[10]=1.0;
inverseconversion[11]=0.01689437;
inverseconversion[12]=0.01224665;
inverseconversion[13]=0.01642202;
inverseconversion[14]=0.0887322;
inverseconversion[15]=0.01031822;
inverseconversion[16]=1.7315543;
inverseconversion[17]=0.0874704;
inverseconversion[18]=0.01193074;
inverseconversion[19]=0.01193074;
inverseconversion[20]=0.0560896;
inverseconversion[21]=0.2394657;
inverseconversion[22]=0.442594;
inverseconversion[23]=0.01193074;
inverseconversion[24]=0.2279358;
inverseconversion[25]=0.013;
inverseconversion[26]=0.01193074;
inverseconversion[27]=192.01361;
inverseconversion[28]=304.37793;
inverseconversion[29]=0.742943;
inverseconversion[30]=0.1310691;
```

```
inverseconversion[31]=1.0709465;
inverseconversion[32]=1.6001887;
inverseconversion[33]=0.0184868;
inverseconversion[34]=0.01993788;
inverseconversion[35]=16.598342;
inverseconversion[36]=0.452371;
inverseconversion[37]=1.2549251;
inverseconversion[38]=0.0246973;
inverseconversion[39]=0.381482;
inverseconversion[40]=0.044981;
inverseconversion[41]=0.0460106;
inverseconversion[42]=0.2279518;
inverseconversion[43]=0.0419028;
inverseconversion[44]=1.5949573;
inverseconversion[45]=0.0771325;
inverseconversion[46]=0.0552406;
inverseconversion[47]=0.694258;
inverseconversion[48]=0.01175709;
inverseconversion[49]=0.02710852;
inverseconversion[50]=0.033366;

boolean found= false;//declaring and intializialing variable
boolean key=false;//declaring and intializialing variable
double c,d;
int pos2=0,pos1=0,repeat=0,repeat1=0;//declaring and intializialing
variable
String cname="";//declaring and intializialing variable
int repeat2=0;//declaring and intializialing variable
double b;//declaring variable
System.out.println("----------------------------------------");
```

```
System.out.println("Enter the name of the country from the list");
int h=0;

for(int k =1;k<=10;k++)//printing country names
{
    for(int p = 0;p<5;p++,h++){
    System.out.print(countryname[h]+",");
 }
 System.out.println();
                          }
System.out.println("----------------------------------------");
Scanner sc =new Scanner(System.in);//Scanner class


while(repeat2==0)//while loop
{
    do//do while loop
    {
       //reinitialisation
       pos1=0;
       pos2=0;
       repeat=0;
       repeat1=0;
       repeat2=0;
       found=false;
       key=false;

          System.out.println("Enter the name of the country whose currency has
to be converted: ");//asking user to input value
           String a=sc.next();//accepting value
```

```
a=a.toLowerCase();


for(int i=0;i<countryname.length;i++)//for loop
{

countryname[i]=countryname[i].toLowerCase();//converting array
element to lower case
    if(a.equals(countryname[i]))//if condition
  {
     found=true;
      pos1=pos1+i;
      repeat=1;
      break;//Break statement
  }
}
    if(found==false)//if condition
   {
      System.out.println("Sorry, the name could not be found.");
      System.out.println("Do you want to repeat?");
      System.out.println("Enter 0 for repeating and any other number if
you want to exit.");
      repeat=sc.nextInt();

   }
}while(repeat==0);


do//do whole loop
{
```

```
            System.out.println("Enter the name of the country to which you want
to convert: ");//asking user to input value
            cname=sc.next();//accepting value
            cname=cname.toLowerCase();

            for(int j=0;j<countryname.length;j++)//for loop
            {

             countryname[j]=countryname[j].toLowerCase();

             if(cname.equals(countryname[j]))
              {
                key=true;
                pos2=pos2+j;
                repeat1=1;
                break;
              }
            }

            if(key==false)//if condition
            {
              System.out.println("The country name is not in the list.");
              System.out.println("Do you want to repeat:");
              System.out.println("Enter 0 for repeating and enter any other
number if you want to exit.");
              repeat1=sc.nextInt();

            }

          }while(repeat1==0);
```

```
        //end of do while loop
        if(found==true&&key==true)
        {
          System.out.println("How many"+" "+currencyname[pos1]+ "  "+
"needs to be converted?");
          b=sc.nextDouble();
          c=b*currencytoIndian[pos1];//currency conversion
          d=c*inverseconversion[pos2];//currency conversion
          System.out.println("The equivalent"+"  "+currencyname[pos2]+"
"+"is"+""+d);
        }
        System.out.println("Do you want to continue the currency
conversion?");
        System.out.println("Enter 0 to continue or any other number to exit.");
        repeat2=sc.nextInt();

      }
      System.out.println("THANKYOU!!!");
      sc.close();//closing Scanner class
   }//end of main
 }//end of class
```

OUTPUT

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Options

```
--------------------------------------
Enter the name of the country from the list
Rwanda,Somalia,Sri Lanka,Sudan,Thailand,
Tonga,Pakistan,Uganda,Uzbekistan,Uruguay,
India,Singapore,Usa,Canada,Denmark,
UK,Japan,China,Holland,Italy,
Sweden,Malaysia,Mexico,Thailand,Germany,
Turkey,Spain,Portugal,Indonesia,Venezuela,
Russia,Afganistan,Nepal,Australia,New Zeland,
South Korea,Ukraine,Bangladesh,Brazil,Taiwan,
UAE,Saudi Arabia,Turkey,Israel,Sudan,
France,Poland,Phillipines,European Union,Fiji,
--------------------------------------
Enter the name of the country whose currency has to be converted:
India
Enter the name of the country to which you want to convert:
USA
How many Rupees  needs to be converted?
1000
The equivalent  American dollar  is12.246649999999999
Do you want to continue the currency conversion?
Enter 0 to continue or any other number to exit.
```

Type input and press Enter to send to program

Type here to search

ENG 00:40 27-11-2022

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Options

```
Russia,Afganistan,Nepal,Australia,New Zeland,
South Korea,Ukraine,Bangladesh,Brazil,Taiwan,
UAE,Saudi Arabia,Turkey,Israel,Sudan,
France,Poland,Phillipines,European Union,Fiji,
---------------------------------------
Enter the name of the country whose currency has to be converted:
India
Enter the name of the country to which you want to convert:
USA
How many Rupees  needs to be converted?
1000
The equivalent  American dollar  is12.246649999999999
Do you want to continue the currency conversion?
Enter 0 to continue or any other number to exit.
0
Enter the name of the country whose currency has to be converted:
Fiji
Enter the name of the country to which you want to convert:
Japan
How many Fijian Dollar  needs to be converted?
160
The equivalent  Japanese Yen  is10219.98533043376
Do you want to continue the currency conversion?
Enter 0 to continue or any other number to exit.
```

Type input and press Enter to send to program

Type here to search

ENG 00:42 27-11-2022

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Options

```
UAE,Saudi Arabia,Turkey,Israel,Sudan,
France,Poland,Phillipines,European Union,Fiji,
--------------------------------------
Enter the name of the country whose currency has to be converted:
India
Enter the name of the country to which you want to convert:
USA
How many Rupees  needs to be converted?
1000
The equivalent  American dollar  is12.246649999999999
Do you want to continue the currency conversion?
Enter 0 to continue or any other number to exit.
0
Enter the name of the country whose currency has to be converted:
Fiji
Enter the name of the country to which you want to convert:
Japan
How many Fijian Dollar  needs to be converted?
160
The equivalent  Japanese Yen  is10219.98533043376
Do you want to continue the currency conversion?
Enter 0 to continue or any other number to exit.
9
THANKYOU!!!
```

Can only enter input while your programming is running

Type here to search

ENG 00:42 27-11-2022

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Options

```
--------------------------------------
Enter the name of the country from the list
Rwanda,Somalia,Sri Lanka,Sudan,Thailand,
Tonga,Pakistan,Uganda,Uzbekistan,Uruguay,
India,Singapore,Usa,Canada,Denmark,
UK,Japan,China,Holland,Italy,
Sweden,Malaysia,Mexico,Thailand,Germany,
Turkey,Spain,Portugal,Indonesia,Venezuela,
Russia,Afganistan,Nepal,Australia,New Zeland,
South Korea,Ukraine,Bangladesh,Brazil,Taiwan,
UAE,Saudi Arabia,Turkey,Israel,Sudan,
France,Poland,Phillipines,European Union,Fiji,
--------------------------------------
Enter the name of the country whose currency has to be converted:
turki
Sorry, the name could not be found.
Do you want to repeat?
Enter 0 for repeating and any other number if you want to exit.
```

Type input and press Enter to send to program

Type here to search

ENG 00:43 27-11-2022

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Options

```
--------------------------------------
Enter the name of the country from the list
Rwanda,Somalia,Sri Lanka,Sudan,Thailand,
Tonga,Pakistan,Uganda,Uzbekistan,Uruguay,
India,Singapore,Usa,Canada,Denmark,
UK,Japan,China,Holland,Italy,
Sweden,Malaysia,Mexico,Thailand,Germany,
Turkey,Spain,Portugal,Indonesia,Venezuela,
Russia,Afganistan,Nepal,Australia,New Zeland,
South Korea,Ukraine,Bangladesh,Brazil,Taiwan,
UAE,Saudi Arabia,Turkey,Israel,Sudan,
France,Poland,Phillipines,European Union,Fiji,
--------------------------------------
Enter the name of the country whose currency has to be converted:
Thailand
Enter the name of the country to which you want to convert:
Mexico
How many Thai baht  needs to be converted?
1209
The equivalent  Thai Baht  is1208.9962322724
Do you want to continue the currency conversion?
Enter 0 to continue or any other number to exit.
```

Type input and press Enter to send to program

Store

Type here to search

ENG 00:45 27-11-2022

VARIABLE DESCRIPTION TABLE

VARIABLE NAME	DATA TYPE	DESCRIPTION
found	boolean	used as a key for searching in array of list of countries
key	boolean	used as a key for searching in array of list of countries
i	int	counter variable
j	int	counter variable
repeat	int	used as key to repeat do while loop
repeat1	int	used as key for while loop
repeat 2	int	used to repeat conversion
pos1	int	to find position of country
pos2	int	to find position of currency
cname	string	to accept country name from user.
b	double	to accept amount to be converted
c	double	to convert entered currency to Indian currency
d	double	to convert Indian currency to destination currency.

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